

ALI ALFRIDAWI

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EDUCATION

University of Texas at Arlington

GPA: 4.0/4.0

Honors Bachelor of Science in Electrical Engineering and Mathematics

Aug. 2025 – May 2029

Honors: Nokia Outstanding CS Student, John A. Gardner, President's List, Maverick Academic Scholarship

Coursework: Electronics, Digital Circuits & Systems, Circuit Analysis, Active & Passive Devices, C Programming

TECHNICAL SKILLS

Languages: C/C++, Go, Lua, Python, Java, JavaScript, HTML/CSS

Frameworks & Libraries: FastAPI, React, Angular, Node.js, Flask, PySide6, Pandas, NumPy

Databases: PostgreSQL, MySQL, SQLite, Redis

Developer Tools: Linux, Docker, Prometheus, Grafana, Git, GitHub, Caddy, systemd, SNMP

EXPERIENCE

Software Engineering Intern

May 2026 – Present

IBM

Houston, TX

- Engineered a full-stack Go, Prometheus, and Docker monitoring system, reducing undetected lab failure by **90%**
- Developed custom SSH and SNMP exporters to continuously collect real-time telemetry from specialized hardware
- Deployed the monitoring infrastructure on a custom VM using systemd and enterprise auth to ensure high availability

Undergraduate Research Assistant

Dec. 2025 – May 2026

University of Texas at Arlington

Arlington, TX

- Developed **Python** simulation scripts to model optical properties and guide experimental design decisions for PCS
- Synthesized complex findings from cutting-edge research papers into digestible weekly briefings for the Zhou Group
- Built a PySide6 graphical user interface for simulation workflows, increasing simulation efficiency by **50%**

Information Technology Intern

Sep. 2024 – May 2025

iFortriss

Plano, TX

- Architected and deployed **2** small office network infrastructures, ensuring secure and stable connectivity
- Delivered responsive and user-focused support, resolving issues while ensuring a positive experience for clients
- Developed an automated inventory tracking system to replace manual data entry, decreasing tracking time by **95%**

PROJECTS

FaultScope | *C++, Go, Linux, Docker, Prometheus*

- Built a distributed monitoring system in C++ and Go that collects telemetry and produces incident timelines
- Improved system reliability during outages by adding local data buffering and duplicate-safe ingestion
- Processed **31,973 samples per second** with zero errors across a **100,000-sample** performance test

Fluurish | FidHacks Winner | *FastAPI, PostgreSQL, Docker, Gemini, React Native*

- Architected a FastAPI and PostgreSQL backend for a financial-literacy app for learning, budgeting, and profiles
- Integrated Gemini's vision API to extract and categorize receipt data with validation and fallback handling
- Implemented Pytest coverage for authentication, persistence, quiz scoring, rewards, resets, and database migrations

Home Lab | *Docker, Caddy, Prometheus, Grafana, PostgreSQL, Redis, Forgejo*

- Designed a containerized home-lab platform hosting reverse proxy, DNS filtering, and monitoring services
- Configured service health monitoring, infrastructure metrics, container-level telemetry, and alert routing
- Self-hosted Forgejo for Git-based source control and Dozzle for container operations

LEADERSHIP

Research Interest Group | *Society of Asian Scientists and Engineers*

Jul. 2026 – Present

- Leading a 5-member research team through the design and implementation of a Rust-based LevelDB clone

HackUTA Tech Team | *Association of Computational Machinery*

Jul. 2026 – Present

- Member of Technical Staff

Cultural Dinner Chair | *Freshman Leaders on Campus*

Aug. 2025 – May 2026

- Formed and led a **5**-member committee to organize a Cultural Dinner event for **50+** attendees